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Climate change policy, market structure, and carbon leakage

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Abstract

The 1997 Kyoto Protocol on climate change obliges the industrialized countries to initiate the international effort of abating anthropogenic greenhouse gas (GHG) emissions. If such an initiative is to be taken, the associated competitive effects may lead to significant relocation of developed countries' energy-intensive production. This paper examines this issue. I adopt an oligopolistic structure combined with increasing returns to scale production technologies to represent the strategic interaction among the firms producing energy-intensive products. This representation is then embedded within a multi-regional computable general equilibrium model, which in turn is used for quantifying these relocation effects. The results suggest that significant relocation of energy-intensive industries away from the OECD may occur, depending on the type of market structure, with leakage rates as high as 130%, in which case GHG control policies in the industrialized countries actually lead to higher global emissions.

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1. Introduction

The 1997 Kyoto Protocol obliges developed industrial countries to initiate an international effort to abate anthropogenic greenhouse gas (GHG) emissions ([Conference](#)

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of the Parties to the United Nations Framework Convention on Climate Change, 1998).¹

As only one part of the world would constrain its emissions under this agreement, many concerns are raised about the agreement's effects on trade and the location of industrial production. One particular effect is that the associated competitive effects may lead to significant offsetting emissions increases in the countries without controls—a phenomenon typically referred to in the literature as carbon leakage.² The leakage process may occur through two different but related channels. First, the lower world prices of energy (reduced demand in the constrained economies exerts a downward pressure on energy prices) encourage substitution towards energy in countries without a carbon constraint. Second, energy-intensive industry may relocate from countries with carbon controls to those without, and export its energy-intensive products back to the constrained countries. The existing results on production effects and leakage from a carbon policy in industrialized countries are mainly derived from computable general equilibrium (CGE) models.³ Though these models treat the first channel relatively well, they fall short of adequately representing the industry relocation channel. For example, some of these models do not even have a separate representation of production and trade in energy-intensive products, whereas others, which include such a representation, suffer from at least three maintained assumptions: constant returns to scale (CRTS) technologies, perfect competition, and an Armington structure characterizing trade in energy-intensive products.⁴ Given the restrictive nature of the modeling assumptions, the scope for leakage tends to be quite limited in these models.⁵ This paper addresses these shortcomings by exploring more fully the effects of the market structure on the geographic distribution of energy-intensive production, trade, and leakage. In doing so, the paper captures two sources that may contribute significantly to the offshore production and leakage effect: the pro-competitive effect of carbon abatement policies in an initially monopolized industry, and the effect of entry and exit of firms in such an industry. These effects are captured by explicitly modelling the strategic interaction among firms producing energy-intensive products through spatial Cournot oligopolies with free entry and exit⁶ in a seven-commodity seven-region applied general equilibrium model of the world economy. With this richer

¹ In the Kyoto Protocol the OECD as defined in 1990, the economies of the Former Soviet Union and the East European economies in transition agreed to take on reduction commitments of a level on average about 5% below their 1990 greenhouse gas emissions levels. The list of these countries was included in Annex B of the Kyoto Protocol and are therefore often referred to as “Annex B countries”.

² Under subglobal abatement, the carbon leakage rate in a non-abating region is defined as the change in its emissions as a fraction of the emissions reduction by the abating regions. The global leakage rate is, then, simply the sum of the regional leakage rates.

³ Examples are Babiker et al. (1997), Felder and Rutherford (1993), and OECD (1992). For a brief survey of CGE models used to study the economic effects of climate change (including leakage) see Gaskin and Weyant (1993) and Weyant (1999).

⁴ It is well recognized that energy-intensive industries such as iron and steel, chemical products, paper and pulp are characterized by economies of scale and market power (Markusen et al., 1993).

⁵ The estimates of leakage reported in these studies range between 6% and 25%.

⁶ Spatial competition in Cournot oligopolies has been used extensively in the IO and the International trade literature for modeling firm location and in studying market structures (e.g. Greenhut and Greenhut, 1975; Brander, 1981; Markusen, 1981; Brander and Krugman, 1983; Venables, 1985; Markusen and Venables, 1988).

representation of the leakage process, we find cases where significant relocation of energy intensive industries away from OECD occurs with leakage rates greater than 100%.

Whether the Kyoto Protocol will go into force (or not) remains an open question, given the recent deadlock in international climate negotiations (Jacoby and Reiner, 2001) and the latter developments in Bonn and Marrakech (Babiker et al., 2002). Nevertheless, the general results of this paper are readily applicable to any attempt to implement climate policies in which a significant portion of the world is not included.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 presents the numerical model. Section 4 describes calibration and baseline. Section 5 presents the results, and Section 6 provides some concluding remarks.

2. Data

The study makes use of a comprehensive energy-economy dataset that accommodates a consistent representation of energy markets in physical units as well as detailed accounts of regional production and bilateral trade flows for the year 1992. This hybrid dataset is based on the Global Trade Analysis Project (GTAP) data along with the International Energy Agency (IEA) satellite accounts on energy balances, prices, and energy taxes. The GTAP database provides consistent global accounts of production, consumption, and bilateral trade. Version 3 of the database covers 30 countries and regions and 37 commodity classifications. The base year for the accounts in version 3 is 1992. The IEA annual energy balances cover most countries, but the regional coverage for the corresponding price and tax data is limited. The details of combining the GTAP and IEA data to construct the dataset used in this study are documented in Rutherford and Babiker (1997). For the purposes of this study, the dataset is aggregated into seven regions and seven commodity groups. The regions are the OECD as defined in 1990 (OEC), oil-exporting countries (OEX), Former Soviet Union (FSU), China and India (CHI), the dynamic Asian economies (DAS), the dynamic economies of South America (DSM), and the rest of the world (ROW). The commodities include five energy goods and two non-energy composites. The energy goods identified in the study are coal (COL), natural gas (GAS), crude oil (CRU), refined oil (OIL), and electricity (ELE), a disaggregation that is essential to distinguish energy goods by their carbon intensity and degree of substitutability. The two non-energy goods are an energy-intensive tradable product (EIS) and a non-energy-intensive tradable composite (OTH). The concordance between the GTAP region and commodity classifications and those adopted in this study is provided in Table 1. The primary factors in the dataset are labor, capital, and fossil-fuel resources. Labor and capital represent gross earnings denominated in 1992 US dollars. The gross returns on the ownership of fossil-fuel resources are imputed from the capital earnings according to a procedure that will be described later in the data calibration section.

Some of the data features that are useful for understanding the results of this paper are displayed in Tables 2 and 3, and Fig. 1. Table 2 displays the value added statistics for the EIS sector and the total value added (GDP) across regions. Note the relatively higher contributions of the EIS industry in regions DAS, DSM, and OEC compared to regions OEX and CHN. Table 3 shows production, consumption and bilateral trade in

Table 1
Regions, commodity classifications and mappings in the model

GTAP region	Aggregated region	GTAP commodity	Aggregated commodity
Australia	OEC	Paddy rice	OTH
New Zealand	OEC	Wheat	OTH
Japan	OEC	Grains (other than rice and wheat)	OTH
Republic of Korea	DAS	Non-grain crops	OTH
Indonesia	OEX	Wool	OTH
Malaysia	DAS	Other livestock	OTH
Philippines	DAS	Forestry	OTH
Singapore	DAS	Fishing	OTH
Thailand	DAS	Coal	COL
China	CHI	Crude oil	CRU
Hong Kong	DAS	Natural gas	GAS
Taiwan	DAS	Other minerals	OTH
India	CHI	Processed rice	OTH
Rest of South Asia	ROW	Meat products	OTH
Canada	OEC	Milk products	OTH
United States of America	OEC	Other food products	OTH
Mexico	OEX	Beverages and tobacco	OTH
Central America and Caribbean	ROW	Textiles	OTH
Argentina	DSM	Wearing apparel	OTH
Brazil	DSM	Leather goods	OTH
Chile	DSM	Lumber and wood	OTH
Rest of South America	ROW	Pulp and paper	EIS
European Union 12	OEC	Petroleum and coal products	OIL
Austria Finland and Sweden	OEC	Chemicals, rubber and plastics	EIS
European Free Trade Area	ROW	Non-metallic mineral products	EIS
Central European Associates	ROW	Primary ferrous metals	EIS
Former Soviet Union	FSU	Non-ferrous metals	EIS
Middle East North and Africa	OEX	Fabricated metal products	EIS
Sub-Saharan Africa	ROW	Transport industries	OTH
Rest of world	ROW	Machinery and equipment	OTH
		Other manufacturing products	OTH
		Electricity water and gas	ELE
		Construction	OTH
		Trade and transport	OTH
		Other services (private)	OTH
		Other services (public)	OTH
		Dwellings	OTH

energy intensive products in 1992. In this table, the rows display the allocation of production to domestic markets and exports and the columns display the composition of consumption from domestic markets and imports. The most noticeable feature of the table is the global concentration of energy-intensive production and consumption, with the OECD producing about 77% and consuming about 75% of the world's energy-intensive products in 1992. Another important feature is the larger trade flows between OECD and non-OECD countries, compared to those among non-OECD countries. [Fig. 1](#)

Table 2

Value added in 1992 (billion US\$)

	Aggregate	EIS only	EIS % share
OEC	16,805	1504	9
FSU	487	36	7
CHI	524	21	4
OEX	1039	61	6
DSM	597	68	11
DAS	750	71	10
ROW	1381	92	7

compares energy intensities across the different regions in the dataset. Two features of the graph worth noting are the larger energy value shares in the energy-intensive sector compared to the national economy, and the higher sectoral as well as aggregate energy intensities in non-OECD economies (particularly those of regions CHN and FSU) compared to OECD.

3. The numerical model

The modeling framework adopted in this analysis is a multi-commodity multi-region static CGE model of the world economy. The key features of the model are briefly outlined below.

3.1. Production technologies

There are three types of production activities in the model: those of fossil fuels (CRU, COL, OIL, and GAS), those of ELE and OTH, and those of the energy-intensive good EIS. Fossil fuels are produced subject to decreasing returns to scale (except refined oil, which is characterized by CRTS) and are traded in perfectly competitive markets. ELE and OTH are produced by perfectly competitive firms according to CRTS technologies. EIS is

Table 3

Production, consumption and bilateral trade in energy-intensive products (1992) (billion US\$)

	OEC	FSU	CHI	OEX	DSM	DAS	ROW	PROD
OEC	3880.1	4.1	16.9	42.8	6.9	49.6	65.3	4065.7
FSU	7.2	100.4	2.0	0.2	0.1	0.8	1.6	112.3
CHI	6.9	0.3	137.9	1.2	0.1	4.8	1.8	153.0
OEX	15.0	0	1.7	150.4	0.7	2.7	2.6	173.1
DSM	8.8	0	0.8	0.9	190.3	2.1	2.6	205.5
DAS	22.4	0.1	7.7	3.1	0.1	237.9	3.0	274.3
ROW	50.3	0.8	1.7	4.2	1.3	4.3	265.2	327.8
CONS	3990.7	105.7	168.7	202.8	199.5	302.2	342.1	5311.7

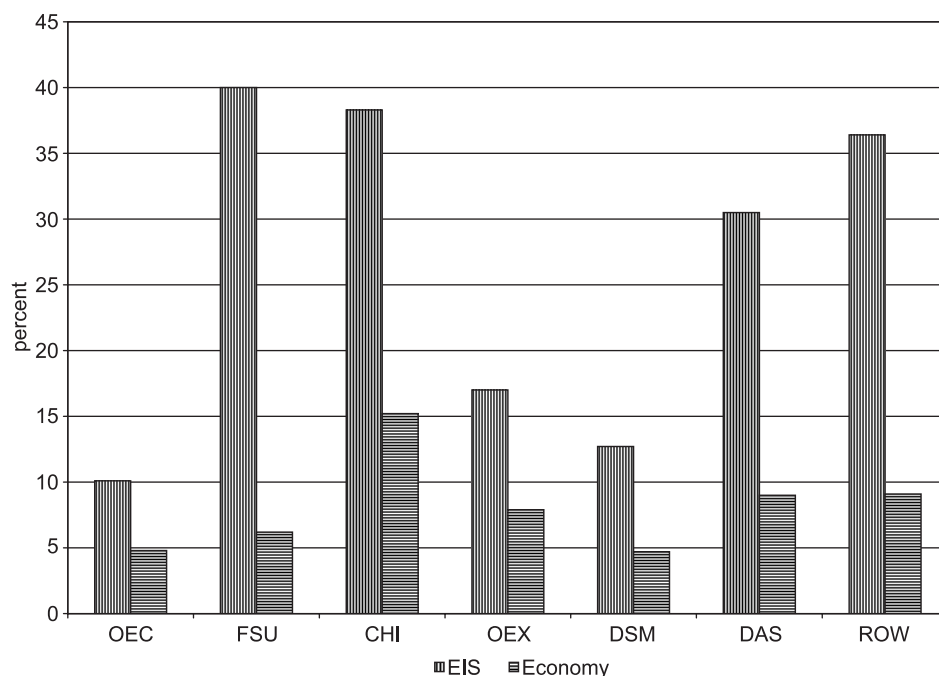


Fig. 1. Energy value shares in 1992.

produced by imperfectly competitive firms under conditions of fixed costs and increasing returns to scale (IRTS).⁷

For all goods, an index Y_{ir} denotes the level of production of good i in region r (for the EIS: $Y_{ir} = n_r y_{ir}$, where n_r is the number of active firms and y_{ir} is the output per firm). Except for crude oil, which is modeled as a perfectly homogenous good, good i is produced as a differentiated product for sale in domestic and international markets. The shares of sales at home and abroad are determined by relative prices. The allocation of sales for each tradable good between domestic and exports is characterized by a constant-elasticity-of-transformation (CET) function.⁸

$$Y_{ir} = [a_{ir}D_{ir}^\eta + b_{ir}X_{ir}^\eta]^{1/\eta} \quad (1)$$

where a and b are share coefficients and the elasticity of transformation between domestic and exports is given by $\sigma^{\text{tr}} = 1/(1 + \eta)$.

To describe the input mix, let N denote the set of non-fossil fuel goods (i.e. ELE, OTH, and EIS), f the set of fossil fuel goods (i.e. CRU, COL, GAS, OIL), and e the set of all energy goods (i.e. CRU, COL, GAS, OIL, and ELE). The production structure for

⁷ In terms of variable inputs, the technology producing EIS is CRTS but, because there are sunk costs, the overall production activity is therefore displays IRTS.

⁸ A representation in which the energy-intensive tradable good is homogenous is also considered in the study.

the non-fossil good N is depicted on subfigure (a) of Fig. A-1 in which the technology producing good N is represented by a nested CES function based on non-energy intermediates, Z , an energy component, E , and a primary factor composite, V .

$$Y_{Nr} = \min[\min_{i \notin e}(\delta_{iNr} Z_{iNr}), (\beta_{Nr} E_{Nr}^{\rho^E} + \gamma_{Nr} V_{Nr}^{\rho^E})^{1/\rho^E}] \quad (2)$$

where δ , β and γ are share coefficients of the CES function and $\sigma^E = 1/(1 - \rho^E)$ is the elasticity of substitution between the energy and the primary-factors composites. The energy composite is a nested CES function combining refined oil, gas, coal, and electricity. The primary factor composite is a Cobb–Douglas aggregate of labor (L) and capital (K):

$$V_{Nr} = L_{Nr}^{\omega_{Nr}} K_{Nr}^{1-\omega_{Nr}} \quad (3)$$

where ω is the labor share.

The production structure of fossil fuels is depicted on subfigure (b) of Fig. A-1. Fossil fuel f is produced according to a nested CES function combining a fuel-specific resource (R), labor, and intermediate inputs.

$$Y_{fr} = [\alpha_{fr} R_{fr}^{\rho_{fr}^R} + v_{fr} (\min(Z_{ifr}, L_{fr}))^{\rho_{fr}^R}]^{1/\rho_{fr}^R} \quad (4)$$

where α and v are the share coefficients of the CES. In this expression production is characterized by the presence of a “a fixed factor” resource (R) that trades off against the rest of inputs in the top level nest according to the substitution elasticity $\sigma_{fr}^R = 1/(1 - \rho_{fr}^R)$, and each of the rest of the inputs enters in fixed proportion in the second level nest.

3.2. Consumer preferences

In each region, r , the preferences of the representative consumer are represented by the nested Constant-Elasticity-of-Substitution (CES) utility function:

$$U_r = \left[\mu_r \left(\prod_{i \in e} C_{ir}^{\theta_i} \right)^{\rho^E} + v_r \left(\prod_{i \notin e} C_{ir}^{\theta_i} \right)^{\rho^E} \right]^{1/\rho^E} \quad (5)$$

where θ is the consumption value share and μ and v are the CES share coefficients. The elasticity of substitution between the energy and non-energy composites is given by σ^E and where each of the two consumption composites has a Cobb–Douglas representation.

3.3. Supplies of final goods and international trade

With the except of crude oil, which is a homogenous good, intermediate and final consumption goods are differentiated following the Armington assumption. For each demand class, the total supply of good i is a CES composite of a domestically produced variety and an imported one.

$$A_{ir} = \sum_j Z_{ijr} = [\psi_{ir}^z Z D_{ir}^{\rho^D} + \zeta_{ir}^z Z M_{ir}^{\rho^D}]^{1/\rho^D} \quad (6)$$

$$C_{ir} = [\psi_{ir}^c CD_{ir}^{\rho^D} + \zeta_{ir}^c CM_{ir}^{\rho^D}]^{1/\rho^D} \quad (7)$$

$$G_{ir} = [\psi_{ir}^g GD_{ir}^{\rho^D} + \zeta_{ir}^g GM_{ir}^{\rho^D}]^{1/\rho^D} \quad (8)$$

where A , Z , C , and G are, respectively, aggregate demand, inter-industry demand, consumer demand, and government demand of the good in question, and where ZD and ZM , CD and CM , and GD and GM are, respectively, domestic and imported components of the composites Z , C , and G . The ψ s and ζ s are the CES share coefficients and the Armington substitution elasticity between the domestic and the imported variety in these composites is $\sigma^D = 1/(1 - \rho^D)$. On the other hand, in each region, the imported varieties of good i are represented by the CES aggregate:

$$M_{ir} = \left[\sum_s \varphi_{isr} x_{isr}^{\rho^M} \right]^{1/\rho^M} \quad (9)$$

where M_{ir} is total imports of good i in region r , x_{isr} is the amount of the good imported from region s , φ_s are the CES share coefficients, and $\sigma^M = 1/(1 - \rho^M)$ is the implied substitution elasticity across origins.

3.4. Emissions

Carbon emissions are generated according to the stoichiometry of fossil fuel combustion, which occurs in fixed proportions to the consumption of fossil fuels by industry and final demand sectors. The carbon emissions in region r are defined by the expression:

$$\text{Emissions}_r = \sum_f \text{Co2Coef}_f \left(\sum_i Z_{fir} + C_{fr} \right) \quad (10)$$

in which Co2Coef is the carbon content (expressed in heat unit of tons/Exajoule).⁹ Given the detailed energy representation in the model, carbon abatement may be achieved either by reducing the amount of energy per unit of output or consumption, or by changing the fossil-fuel mix.

3.5. Optimization behavior and market structure

In the standard manner, the representative consumer chooses C_{ir} to maximize the utility function in Eq. (5) subject to a budget constraint that consumption expenditure equals income:

$$\max_{C_{ir}} U_r(C_{ir}) \text{ s.t. } \sum_i p_{ir}^c C_{ir} = p_r^k K 0_r + p_r^l L 0_r + \sum_f p_{fr}^R R 0_{fr} + \text{TR}_r - \sum_i p_{ir}^g G_{ir} - p_r^I I_r \quad (11)$$

⁹ The ratio of carbon in coal/oil/gas is 1:0.75:0.57.

where p^c , p^k , p^l , p^R , p^g , and p^I are price indices for household consumption (gross of taxes and markups), capital services, labor services, rents on fossil-fuel resources, government consumption (gross of taxes and markups), and new vintage capital (gross of taxes and markup), respectively. $K0$, $L0$, and $R0$ are the benchmark stocks of capital, labor, and fossil fuel resources; TR is total tax revenue collected by government, G is government expenditure, and I is investment. In this representation, household consumption is financed from factor incomes net of government budget deficit and investment expenditure. These factor incomes are labor income, capital income, and rents accrued on ownership of fossil-fuel resources. In the model fossil-fuel resources are treated as sector-specific, whereas labor and capital are treated as perfectly mobile across sectors within the same region but immobile across regions. All factors are inelastically supplied. Taxes apply to energy as well as non-energy demand, to production and income, and to international trade. Both the level of government consumption and the level of investment are treated as exogenous variables in the model.

Given input prices gross of taxes and markups, firms maximize profits subject to the technology constraints in Eqs. (1)–(4). Minimizing the input costs for a unit value of output yields a unit cost index (marginal cost), p_{ir}^y . Firms in the perfectly competitive sectors maximize profits by selling their products at a price equal to this marginal cost. In contrast, firms producing in the oligopolistic sector (EIS) maximize profits by selling their output at a price higher than its marginal cost, p_{ir}^y . Given its conjectures about the behavior of other firms in domestic and foreign markets, the representative firm in the EIS industry of region r chooses its supplies to the different markets in such a way that the sum of its profits net of fixed costs is maximized. In the general case of segmented markets and differentiated goods, this problem has the form:

$$\max_{d_r, x_{rs}} \pi_r = [p_r^d(D_r) - p_r^y]d_r + \sum_s [p_{rs}^x(X_{rs}) - p_r^y]x_{rs} - \Gamma_r \quad (12)$$

where d_r and x_{rs} are the firm's sales in the domestic and foreign markets, respectively. $p_r^d(D_r)$ and $p_{rs}^x(X_{rs})$ are the corresponding inverse Marshallian demand functions facing the firm in these markets, with D_r and X_{rs} being the aggregate demands and where Γ_r denotes the fixed production costs incurred by the firm.¹⁰

With Cournot conjectures, the solution to this profit maximization problem yields the optimal markup formulas on domestic and foreign sales:¹¹

$$p_r^d \left[1 - \frac{\phi_r^d}{n_r \epsilon_r^d} \right] = p_r^y \text{ and } p_{rs}^x \left[1 - \frac{\phi_r^x}{n_r \epsilon_{rs}^x} \right] = p_r^y \quad (13)$$

where ϕ_r^d and ϕ_{rs}^x are respectively the industry's shares in the domestic and foreign markets, ϵ is the positive of the Marshallian elasticity of demand, and n_r is the number of

¹⁰ Note that the commodity index in the equation is omitted for transparency.

¹¹ For computational purposes, it is more convenient to specify the markups as ad valorem additions to marginal costs.

active firms in region r . With free entry and exit, firms in the oligopolistic industry earn zero economic profits in equilibrium, i.e.

$$p_r^d \frac{\phi_r^d}{n_r \epsilon_r^d} + \sum_s p_{rs}^x \frac{\phi_r^x}{n_r \epsilon_{rs}^x} x_{rs} \leq \Gamma_r \quad (14)$$

a condition that the sum of markup revenues from domestic and foreign sales should not exceed the fixed production costs incurred by the firm. In equilibrium this condition is satisfied with equality for all operating firms and therefore uniquely determines the number of active firms in each region r . The calibration of the initial markups and fixed costs in the model will be considered in Section 4.

3.6. Market clearance, model closures, and model solution

Having outlined the zero profit equations and the income–expenditure balance, to close the model one needs clearance equations for factor markets, product markets, and foreign trade. Since all the factors in the model are supplied inelastically, factor market clearance conditions are trivial. The other market clearance equations are as follow:

1. Supplies to the domestic market, net of the exogenously determined investment, must meet demands by industry, household and government:

$$D_{ir} = ZD_{ir} + CD_{ir} + GD_{ir} + \text{invest}_{ir} \text{ where } \sum_i \text{invest}_{ir} = \bar{I}_r \quad (15)$$

2. Regional export supplies equal the corresponding regional import demands:

$$X_{ir} = \sum_s x_{irs} \quad (16)$$

3. Import supply of good i satisfies domestic demands by household, government, and industry for the imported commodity:

$$M_{ir} = ZM_{ir} + CM_{ir} + GM_{ir} \quad (17)$$

4. International trade in each commodity has to balance:

$$\sum_r X_{ir} = \sum_r M_{ir} \quad (18)$$

Numerically, the model is formulated and solved as a mixed complementarity problem (MCP) using the GAMS/MPSGE system described in [Rutherford \(1995\)](#).¹²

¹² The complete code is available from the author at http://www.mit.edu/babiker/www/jie_01/jie_01.html.

4. Elasticities, calibration and baseline

4.1. Elasticities

Estimates of the reference values of the elasticities used in the model are obtained from Nainar (1989), Nguyen (1986), Pindyck (1979) and from other CGE models (Gaskin and Weyant, 1993). Table 4 shows these elasticity values.

4.2. Calibration

4.2.1. Fossil fuel resource supplies

Version 3 of the GTAP database does not provide separately exhaustible resource supplies in the fossil fuels sector. Instead, the rental value of the resource is lumped together with the rental value of capital. Given our exogenous assumptions on the price elasticity of supply for fuel, and the substitution elasticity between the exhaustible resource and the other inputs in the production of the fuel (see Table 4), the benchmark share of the resource is calibrated according to the following procedure.

Fossil fuel production levels are determined by the price of fuel relative to the price of domestic output. The production of fuel f requires inputs of domestic supply (e.g. labor and intermediate inputs) and a fuel-specific resource. Given the form of the production function in Eq. (4), the elasticity of substitution between the resource and the rest of inputs in the top nest determines the price elasticity of supply (ξ_f) at the reference point, according to the relation:

$$\xi_f = \sigma_{fr}^R \frac{1 - \alpha_{fr}}{\alpha_{fr}} \quad (19)$$

where α is the resource value share. The imputed returns to the exhaustible resource from this procedure are then netted out from the rental value of capital input in the database.

Table 4
Key elasticities in the model

Elasticity	Description	Value	Comments
σ	Top level elasticity in production	0	Armington structure
	production	1	Heckscher–Ohlin structure
σ^{tr}	The transformation elasticity	2	Segmented markets
		∞	Integrated markets
σ^{E}	Substitution between energy and non-energy in demand	0.5	Armington structure
		1	Heckscher–Ohlin structure
σ^{D}	Armington elasticity between domestic and imports	8	Armington structure
		∞	Heckscher–Ohlin (EIS)
σ^{M}	Armington elasticity across imports	16	Armington structure
		∞	Heckscher–Ohlin (EIS)
ξ_f	Fossil fuel supply elasticity	1	Crude oil
		0.5	Coal
		1	Natural gas

4.2.2. Benchmark markups, fixed costs, and number of firms in the energy intensive industry

The oligopoly markup formulas of Section 3.5 imply the general form for the benchmark markup rate (on cost):

$$MK0 = \frac{\phi0}{n0\epsilon - \phi0} mc0 \quad (20)$$

where $n0$ is the initial number of operating firms, $mc0$ is the benchmark marginal cost, $\phi0$ is the benchmark market share, and ϵ is the uncompensated elasticity of demand. For each region there is one such equation for the domestic market and one such equation for each foreign market. The value of ϵ is inferred from the domestic and the import substitution elasticities of Section 3.3 according to the relation:

$$\epsilon = \sigma - \lambda(\sigma - 1) \quad (21)$$

in which λ is the demand share of the good in question, and σ is the substitution elasticity. In the case where goods are both nationally and regionally differentiated, the substitution elasticity is given by σ^D for the domestic markup and by σ^M for the foreign markups. With the assumption of free entry and exit, the sum of the benchmark markup revenues equals the firm's benchmark fixed costs:

$$MK0 = MKD0_r d0_r + \sum_s MKF0_{rs} x0_{rs} = \Gamma0_r \quad (22)$$

where $d0$ and $x0$ are, respectively, domestic and foreign sales of the representative firm. $MKD0$ is the benchmark domestic markup and $MKF0$ is the foreign markup.

Now, using the benchmark data on production costs, final demands, domestic sales, foreign sales, along with the substitution elasticities and a benchmark 15%¹³ markup on domestic sales, the calibration system is solved recursively for $n0$, $MKF0$, and $\Gamma0$. The fixed costs are constructed by assuming that 75% are specialized sector-specific capital (e.g. patent, R&D, property rights, etc.) and 25% are labor (e.g. management and human capital). The two components of inputs to fixed costs are assumed to substitute for each other with a substitution elasticity of 0.5. To simulate the offshore licensing aspect of industry relocation it is further assumed that the services of the sector-specific capital component are internationally mobile.

4.3. The baseline (business as usual, BAU)

The base year for our database is 1992. However, the first commitment period under the Kyoto Protocol, whose emissions targets are used to derive the policy experiments in this paper, is centered around the year 2010. The baseline for 2010 is constructed by applying the US Department of Energy (DOE) projections of regional economic growth, emissions growth, and world energy prices (U.S. Department of Energy Information Administration,

¹³ For OECD the 15% is rather conservative compared to the average markups reported in the literature (Allayannis and Weston, 1999). Provided the leverage of the markup on the size of pro-competitive effects, with this calibration, the paper's results on production and leakage effects are more in the nature of lower bounds.

Table 5a

Baseline assumption: GDP growth rates (% per annum)

	1992–2000	2000–2005	2005–2010
OEC	2.8	2.3	2.2
DAS	5.8	4.0	4.0
CHI	7.0	5.0	5.0
OEX	3.6	3.7	4.0
DSM	5.6	4.0	4.0
FSU	–2.0	2.0	5.0
ROW	2.5	3.0	3.0

Sources: 1992–2000 [International Monetary Fund \(1998\)](#), 2001–2015 [U.S. Department of Energy, Energy Information Administration, EIA \(1998\)](#).

Table 5b

Baseline assumption: crude oil price (US\$ per barrel)

1995	2000	2005	2010
18.4	19.11	20.2	20.81

Source: [U.S. Department of Energy, Energy Information Administration, EIA \(1998\)](#).

Table 5c

Baseline assumption: autonomous energy efficiency growth rates (1992–2010)

	OEC	FSU	CHI	OEX	DSM	DAS	ROW
COL	26	51	44	31	44	6	44
GAS	0	3	0	0	0	0	0
OIL	17	44	0	18	0	2	0

[EIA, 1998](#)).¹⁴ [Tables 5a–5c](#) report these assumptions. The GDP growth projections and crude oil price forecasts used are shown in [Tables 5a and 5b](#), respectively. The ratio of the GDP growth rates for 2010 to the corresponding CO₂ emissions growth rates for 2010 from DOE is used to account for the non-price induced increases in the efficiency of energy use, generally referred to as autonomous energy efficiency improvement (AEEI). The implied annual rates of AEEI growth thus calculated are shown in [Table 5c](#). This latter table suggests that the bulk of the improvements in energy efficiency is expected to occur in coal burning technologies.

[Table 6](#) shows the evolution of CO₂ emissions in 2010 compared to 1992 and contrasts baseline emissions in the present study with DOE projections for the regions for which DOE has a breakdown. It is interesting that my CO₂ projections (column 2010) compare very well with those of the DOE (column 2010 (DOE)) even though the two set of projections come from quite different models, an indication that economic growth and AEEI are the main driving forces of emissions growth. With respect to regional emissions growth, the table suggests that during the next decade the fastest growth in emissions

¹⁴ A glossary of all data and parameter sources in the study is shown in an [Appendix table](#).

Table 6

Baseline carbon emissions (MMT)

	1992	2010	2010 (DOE)	Kyoto target
OECD	2737	3480	3535	2519
FSU	797	637	666	917
CHI	827	1764	1777	–
OEX	328	528	–	–
DSM	93	192	–	–
DAS	204	457	–	–
ROW	773	968	–	–
TOT	5761	8027	8018	–

(greater than 100%) is expected to occur in China, India, and the dynamic Asian economies. As far as Kyoto emissions targets are concerned, Table 6 indicates that the target for region FSU actually exceeds our projected emissions for the region (column 2010) in 2010 by 280 mmt (million metric tons), the difference is a phenomenon referred to in the climate change literature as “hot air”. In contrast, the table suggests that 28% reductions in CO₂ emissions in 2010 from the baseline are needed in the OECD to meet the Kyoto target for the region.

Focusing on the energy intensive industry (the sector of interest in this study), Table 7 shows the projected production, consumption and bilateral trade in energy intensive products in 2010. As in Table 3, the rows display the allocation of production to domestic markets and exports and the columns display the composition of consumption from domestic markets and imports. Though still the energy-intensive industry continues to be concentrated in the OECD, with the OECD region consuming 68% and producing 69% of the world’s supply, there is a remarkable increase in the production, consumption and trade in energy-intensive products in regions CHI and DAS.

The baseline carbon intensities in the energy-intensive sector and those for the economy as a whole are depicted in Fig. 2. Consistent with the pattern of energy value shares from Fig. 1 the two important features of Fig. 2 are the larger carbon intensities in energy-intensive sectors compared to aggregate regional carbon intensities, and the higher sectoral as well as economy-wide carbon intensities in non-OECD economies compared to OECD. In terms of both of these features, the economies of CHN and FSU rank exceptionally high. The main insights based on these features is the likelihood that expansion of

Table 7

Production, consumption and bilateral trade in energy-intensive products (2010) (billion US\$)

	OECD	FSU	CHI	OEX	DSM	DAS	ROW	PROD
OECD	5304.1	4.3	17.7	63.1	9.2	70.6	87.5	5556.5
FSU	7.8	105.6	2.3	0.2	0.2	1.0	1.9	119.0
CHI	36.6	1.5	458.2	8.4	0.6	30.2	10.5	546.1
OEX	20.6	0.0	2.3	278.0	1.2	4.4	3.8	310.4
DSM	14.3	0.0	1.4	1.9	399.5	4.1	4.6	425.7
DAS	39.5	0.1	13.4	6.8	0.3	540.6	5.7	606.5
ROW	69.8	0.9	2.2	6.6	2.0	6.4	396.3	484.2
CONS	5492.7	112.5	497.6	365.0	413.0	657.4	510.2	8048.3

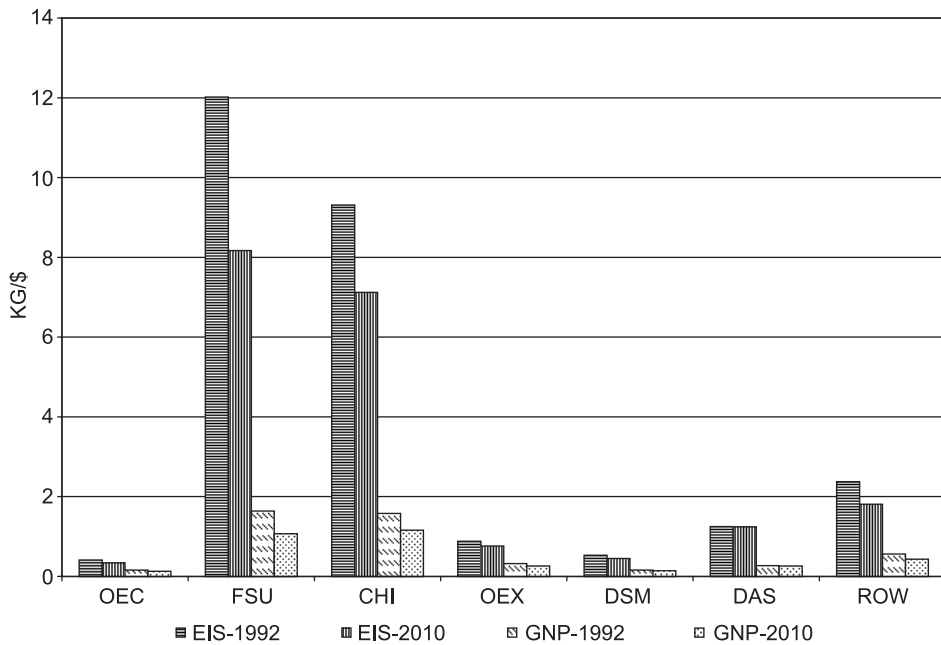


Fig. 2. Baseline carbon intensities.

production of energy-intensive goods in developing countries away from the OECD could lead to a significant increase in global carbon emissions. Significant relocation of production to these regions could thus create significant carbon leakage.

5. Policy cases and result

Without loss of generality, I assume that domestic and export markets are integrated (i.e. $\sigma^{\text{tr}} = \infty$ in Table 4) so that I can focus on differences in trade structure. I distinguish between an Armington case, where the energy-intensive products are both nationally and regionally differentiated, and a Heckscher–Ohlin case,¹⁵ in which the products are perfectly homogenous, maintaining in both cases an increasing returns to scale (IRTS) structure. An intermediate case of a Heckscher–Ohlin model with constant returns to scale (CRTS) is also considered as a yardstick against which to assess the implications of the increasing returns to scale assumption. The elasticity structures corresponding to the Armington and Heckscher–Ohlin cases are described in Table 4. The policy counterfactual considered here is an economy-wide CO₂ constraint in the OECD and FSU to meet the emissions targets agreed to under the Kyoto Protocol. The scenarios considered are *BAU*: the baseline of an Armington model with IRTS, *IRTS-ARM*: Kyoto targets in an

¹⁵ Note that our model is not a full-fledged Heckscher–Ohlin, and the phrase Heckscher–Ohlin is used in a rather restricted sense to indicate that only energy-intensive goods are perfectly homogenous.

Armington model with IRTS, *CRTS-HO*: Kyoto targets in a Heckscher–Ohlin model with CRTS, and *IRTS-HO*: Kyoto targets in a Heckscher–Ohlin model with IRTS.^{16,17} The results are presented in the following order: firm and production effects, welfare effects, effects on trade and offshore licensing of production, and finally effects on carbon leakage.

5.1. Firm and production effects

Table 8 reports the percentage change in the number of firms (concentration) and the percentage change in output per firm (production scale) for the market structures IRTS-ARM and IRTS-HO. In general, the results suggest a remarkable decline in the size of the energy-intensive industry in the OECD and an expansion of the industry outside the OECD. The extent of this structural change, however, is governed by the market structure characterizing production and international trade in energy-intensive products. In particular, the locational effects are smaller when products are differentiated and larger when products are homogenous. The noticeable differences between the results of the two cases reflect the significance of product differentiability among internationally traded goods. With the energy intensive products being considered as perfect substitutes in all regions, both the domestic and the foreign markets are exposed to international competition (the pro-competitive effect in the international trade literature). Hence, the perceived demand curves facing the foreign firm becomes flat, markups fall, foreign sales in OECD increase, and accordingly both total production and production per firm expand in non-OECD economies. Simultaneously, firms in the OECD region lose sales in both domestic and foreign markets due to their cost disadvantage in the presence of more competitive markets. This in turns leads to lower firm concentration and smaller output per firm in the OECD's energy-intensive industry. The other interesting feature of the table is the leverage of economies of scale under the IRTS-HO structure. With traded products being perfect substitutes, the presence of unexploited economies of scale in developing countries facilitate significant expansions of the energy-intensive sector there in the case IRTS-HO. As far as region-specific effects are concerned, Table 8 shows a number of interesting features.

For the OECD, the cost of the abatement policy induces exit of firms and reductions in output per firm. Under the assumption of differentiated goods, the number of firms falls by 2% and output per firm by 3.7%. Relaxing the assumption of product differentiation and moving to a Heckscher–Ohlin world, the number of firms falls by 53.3% and output per firm by 57%. For all non-OECD regions in Table 8, output per firm rises more than the number of firms, particularly in the case of homogenous traded goods, indicating the presence of unexploited economies of scale in these regions. For FSU the carbon constraint is non-binding and hence its energy-intensive industry has experienced a horizontal (number of firms) expansion of 3% and 48% and a vertical

¹⁶ In this market structure the relevant substitution elasticities in the markup formulas correspond to those between intermediates and the rest of the inputs in the production function, and those among the non-energy consumer goods in the utility function.

¹⁷ Other intermediate cases were considered and their results found to fall within the range of the three specific cases reported here.

Table 8

Effects on number of firms and production scale in the energy-intensive industry (% change from BAU)

	IRTS-ARM		IRTS-HO	
	No. of firms	Output of firm	No. of firms	Output of firm
OEC	– 2.0	– 3.7	– 53.3	– 57.0
FSU	2.6	4.7	47.9	67.9
CHI	3.5	6.6	11.5	120.0
OEX	5.0	6.0	0.9	0.6
DSM	1.7	1.8	3.8	13.8
DAS	6.1	8.3	27.9	115.9
ROW	7.2	10.9	15.8	22.5

(output per firm) expansion of 5% and 68%, respectively, for the cases IRTS-ARM and IRTS-HO. DSM experiences the smallest firm and output effects, whereas OEX has larger firm effects when the energy intensive product is an internationally differentiated good. Although ROW has the largest firm effects among non-OECD regions under the Armington structure, it ranks only 4th under the Heckscher–Ohlin structure. However, the most striking industry effects that result from the implementation of the Kyoto Protocol occur in China, India, and the dynamic Asian economies. Although these effects are modest under the IRTS-ARM structure (amounting to 4% growth in firms and 7% growth in scale in CHN and 6% growth in firm and 8% growth in scale in DAS), they are 12% and 120% in CHN and 28% and 116% in DAS under the IRTS-HO market structure.

The effects on the aggregate regional production and consumption of energy intensive products are shown in Table 9.¹⁸ The first general feature reflected by the table is the decrease in both the production and consumption of energy intensive products in the OECD and the corresponding increase in these quantities in the non-OECD regions when Kyoto is implemented. The second feature is the large size of these production and consumption effects when the traded energy-intensive goods are homogenous. The third feature is the larger size of global production and consumption of the energy intensive good under Kyoto in the IRTS structure compared to the CRTS structure (columns CRTS-HO and IRTS-HO). This last result is consistent with the standard intuition that the good is under-produced in the Business as Usual case when production takes place under conditions of imperfect competition compared to the situation in which markets are perfectly competitive. Through its effect on markups (the pro-competitive effect), Kyoto induces greater competition in the energy-intensive market and thus (other things remaining equal) will lead to a larger increase in production of the oligopolistic good compared to the case in which the industry is initially perfectly competitive. Nonetheless, under IRTS the geographical distribution of production and consumption is different from that under CRTS. In particular, Table 9 suggests greater concentration of the EIS industry in regions CHN and DAS under *IRTS-HO* compared to *CRTS-HO*. Thus, regions such as OEX and ROW, which might be attractive for the production of

¹⁸ Note that even though both Tables 8 and 9 show the same production effects from Kyoto, they are different in the sense that Table 8 reports the percentage deviations from the baseline, whereas Table 9 reports the actual numbers.

Table 9

Production and consumption of energy-intensive products in 2010 (billion US\$)

	BAU	IRTS-ARM	CRTS-HO	IRTS-HO
<i>(a) Production</i>				
OECD	5556.5	5351.4	2842.9	1139.2
FSU	119	127.2	151.7	72.2
CHI	546.1	599.8	1964.6	4576.1
OEX	310.4	326.4	1530.4	446.3
DSM	425.7	427.1	499.7	516.5
DAS	606.5	645.5	664.1	1820.4
ROW	484.2	532.9	652.5	591.4
TOT	8048.4	8010.3	8305.9	9162.1
<i>(b) Consumption</i>				
OECD	5492.7	5387.8	4655.9	4007.6
FSU	112.5	115.5	135.6	107.9
CHI	497.6	522.5	1034.0	2246.0
OEX	365.0	369.7	841.6	446.3
DSM	413.0	412.7	465.6	474.3
DAS	657.4	675.4	638.7	1305.1
ROW	510.2	526.7	534.5	574.9
TOT	8048.4	8010.3	8305.9	9162.1

the energy-intensive good if the market were perfectly competitive, actually produce and consume less of the good when the market is characterized by increasing returns to scale and imperfect competition.

5.2. Welfare effects

Fig. 3 depicts the welfare impacts of Kyoto under the different market structure assumptions.¹⁹ Focusing on the two main cases *IRTS-ARM* and *IRTS-HO*, the assumption of homogeneity increases the welfare cost of the policy in the OECD, but not by much. At the same time, however, it causes significant welfare gains in non-OECD regions. This outcome is partially due to the large size of the OECD economy compared to the non-OECD economies, so that industry relocation gives rise to a small reduction in OECD welfare but large welfare gains in the non-OECD. The result may also suggest that the offshore effect in the energy-intensive industry is a more of a shift in comparative advantage than a loss in aggregate competitiveness, i.e. the OECD welfare loss from the outward relocation of energy-intensive production is partially offset by an inward relocation of non-energy-intensive production. The welfare losses in regions OEX and ROW in case *IRTS-ARM* are due to terms of trade effects and to the fall in the energy prices caused by the OECD abatement policy (Babiker et al., 2000). Comparing *IRTS-HO* with *CRTS-HO*, imperfect competition and scale econo-

¹⁹ The environmental benefits of emissions abatement are not included in the calculations. Further, even though the direct comparability across cases of OECD welfare effects is complicated by the fact that the global abatement achieved by the policy differs according to market structure and trade regime, the leakage effect has the same direction as the welfare effect, i.e. cases with higher welfare costs in OECD also produce higher leakage rates.

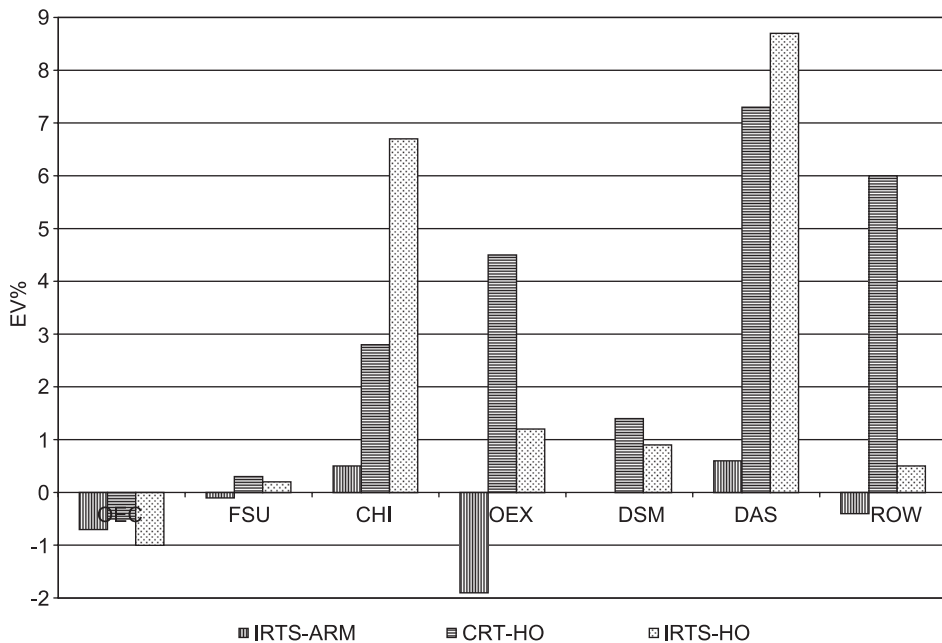


Fig. 3. Welfare effects of Kyoto Protocol.

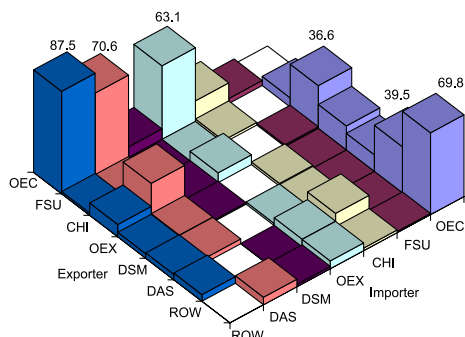
mies seem to affect significantly the distribution of the non-OECD gains conferred by the OECD abatement policy. In particular, the graph indicates that under the homogenous product assumption most of the gains are captured by CHI and DAS when the markets are characterized by IRTS and imperfect competition, whereas a more even distribution of gains emerges if the markets are characterized by CRTS and perfect competition.

5.3. Effects on bilateral trade and offshore licensing

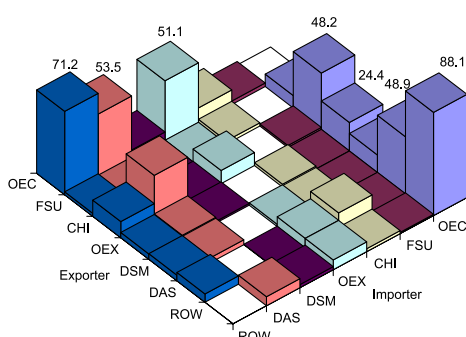
The repercussions of Kyoto on the geographic distribution of international trade in energy-intensive products are portrayed in Fig. 4 for the different market structures.²⁰ Fig. 4a shows that most of the trade in EIS under BAU flows from OECD to non-OECD. The imposition of Kyoto reverses this pattern of trade direction. The extent of this trade reversal, however, depends on the type of market structure. Fig. 4 shows that under product differentiation, there is a noticeable decrease in OECD exports to non-OECD economies and a noticeable increase in OECD imports from these economies. Fig. 4c and d show that under product homogeneity, huge shifts in trade patterns occur between OECD and non-OECD as well as among the non-OECD economies. The differences between Fig. 4c and d, thus, highlight the significance of the IRTS and imperfect competition

²⁰ To facilitate correct interpretation the graph, note that an entry such as that labeled 69.8 in Fig. 4a should be read as “region OEC imports 69.8 in 1992 dollars worth of energy-intensive products from region ROW”.

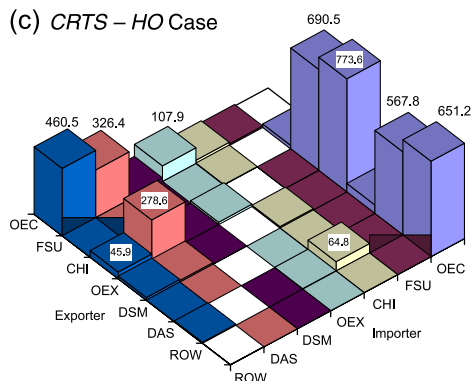
(a) BAU Case



(b) IRTS – ARM Case



(c) CRTS – HO Case



(d) IRTS – HO Case

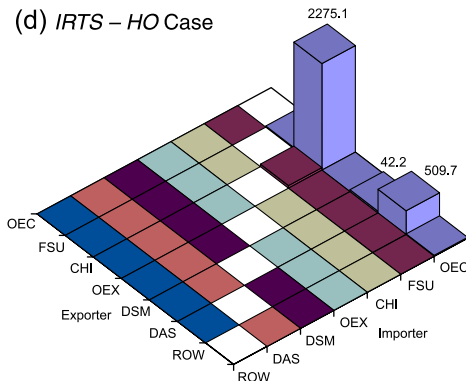


Fig. 4. Bilateral trade in energy intensive products in 2001 (billion 1992 dollars).

assumptions. Whereas the presence of homogeneous products supports bi-directional trade under case CRTS-HO, it eliminates it under case IRTS-HO, leading to a complete specialization of non-OECD economies in exporting the energy intensive products to the OECD, with regions CHI and DAS having the lion share of trade.

The global restructuring of the energy-intensive industry is also manifested in the licensing of offshore production. The implementation of Kyoto results in large flows of overhead capital from the OECD energy-intensive industry to that of the non-OECD economies. The extent of these flows, however, are found to depend very much on the homogeneity of energy-intensive products, with the size of payments for royalties and licensing of patents more than doubles when the market structure is IRTS-HO (about US\$130 billion) compared to the structure IRTS-ARM (about US\$55 billion). As to where the OECD energy-intensive patents flow to, my results suggest that about 70% of the induced FDI occurs in the ROW if the produced products are internationally differentiated, whereas more than 80% ends up in regions CHI and DAS when the produced products are homogeneous.²¹

²¹ This is essentially because the Armington structure preserves the benchmark bilateral trade patterns in Table 3, whereas the Heckscher–Ohlin structure allows for more trade diversification.

5.4. *Effects on carbon leakage*

I conclude the discussion of the effects of the Kyoto Protocol by presenting its environmental impacts. Since the objective of the Protocol is to reduce anthropogenic emissions of GHGs, it is important to assess how much reduction is achieved globally. Fig. 5 shows the carbon leakage rates for the three cases considered in this study. The most striking result is the potential for the global leakage rate to exceed 100%; i.e. a policy to reduce carbon emissions in the industrialized countries actually increases global carbon emissions. This occurs in case IRTS-HO. Although less dramatic, the constant returns to scale case (CRTS-HO) produces a global leakage rate of about 60%, indicating that homogeneity of the traded products is an important source of leakage. This suggests that the presence of unexploited economies of scale in an industry characterized by imperfect competition and product homogeneity is an important source of carbon leakage. In contrast, the differentiated products case (IRTS-ARM) produces a global leakage rate of about only 25%, yet is still significantly above the 20% rate produced by the model for the case of CRTS and differentiated products (not shown), thus, further signifies the leverage of the market structure assumption.

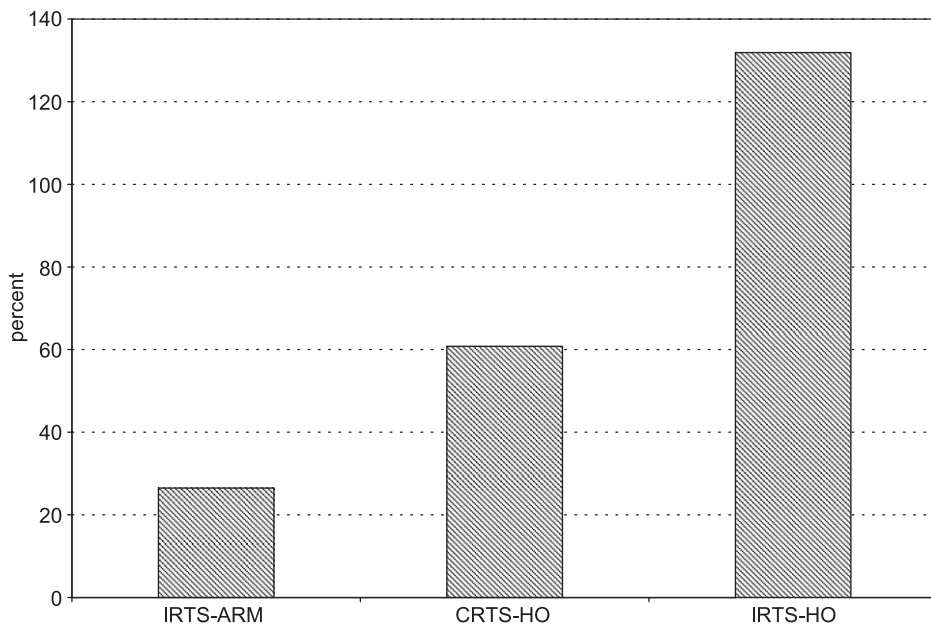
Exploring the geographical details of Fig. 5, the patterns of regional leakage rates are consistent with the pattern of the industry relocation seen in Sections 5.1 and 5.3. In particular, CHI and DAS are the main sources of leakage if products are homogenous, whereas region ROW is the main source if products are differentiated. The most interesting feature, however, is that the expansion of the energy-intensive industry in China and India (CHI) alone might offset 80% of the reductions achieved by the OECD under the Kyoto Protocol if the EIS industry were to exhibit increasing returns to scale combined with imperfect competition and product homogeneity.

6. **Concluding remarks**

Most studies have found that the carbon leakage rate resulting from a policy of the size of the Kyoto Protocol to be in the range 5% to 25%. Yet, all these studies have adopted similar modeling structures that do not capture the phenomenon of industry relocation, particularly over the long run. By extending the structure of these models to include features such as economies of scales, market power, and richer representations of international trade, this paper demonstrates that there exists a wider scope of leakage than previously thought.

The competitive effects induced by OECD abatement of GHGs can lead to significant increases in offshore energy-intensive production associated with relocation of firms. The extent of these locational effects is found to depend crucially on the representation of the market structure and the substitutability among the traded energy-intensive products. When energy intensive products are modeled as Heckscher–Ohlin goods, the global carbon leakage rate is found to range between 50% and 130%, in which case a policy to limit carbon emissions in the OECD has the perverse effect of increasing global emissions. The regional effects of such a policy vary widely, with most of the leakage occurring in China, India, and the dynamic Asian economies.

(a) Global Leakage



(b) Regional Leakage

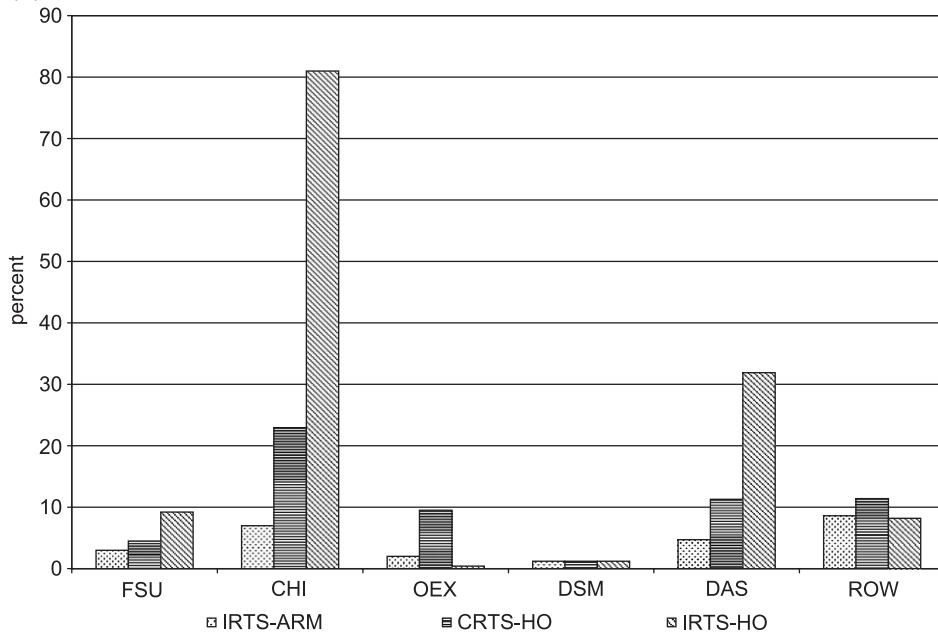


Fig. 5. Leakage effects of the Kyoto Protocol.

These results demonstrate that implementing a policy of limiting carbon emissions that fails to include many regions of the world may, by ignoring the role of the global trading system, fail to achieve its stated ends. The striking results in terms of carbon leakage alone in the Heckscher–Ohlin cases are reasons for the world to pause in implementing subglobal carbon policies if there is a reasonable chance that these model features are realistic. I argue that increasing returns to scale and product homogeneity may not be implausible characteristics of the energy-intensive industry in the long run. First, energy intensive industries such as chemical and petroleum products are recognized to have large economies of scale. Second, technology diffusion through multinational corporations, harmonization of international trade, and the extending international web of communication are all forces that tend to gradually diminish price differences among regions. Hence, a Heckscher–Ohlin assumption is a reasonable or at least not an inconceivable approximation in the long run.

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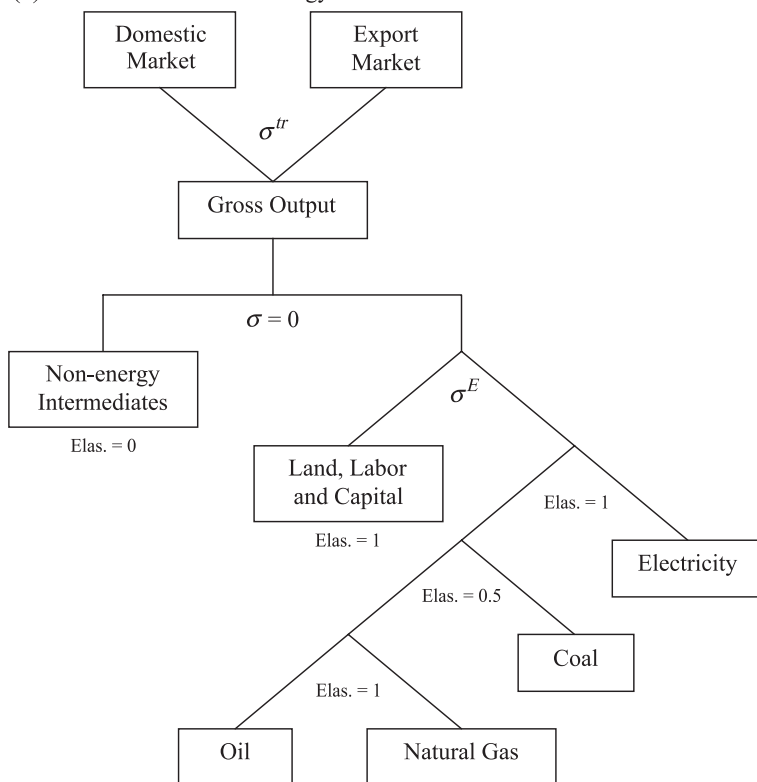
Appendix A

Table A-1

Data Sources

Data and parameters	Source
Social accounting matrices bilateral trade	Global Trade Analysis Project (GTAP) http://www.agecon.purdue.edu/gtap
Physical energy flows, energy prices and energy taxes	International energy agency, IEA Energy Balances, Paris
Projections of fossil-fuel prices and energy efficiency	Energy Outlook, Energy Information Administration (EIA), DOE, USA
Projections of economic growth	Energy Information Administration (EIA) International Monetary Fund (IMF)
Markups in the energy-intensive industry	Allayannis and Weston (1999)
Trade elasticities	GTAP
Energy demand and supply elasticities	Pindyck (1979) , Nguyen (1986) , Nainar (1989) , and Gaskin and Weyant (1993)

(a) Non-Fossil Fuel Technology



(b) Fossil Fuel Technology

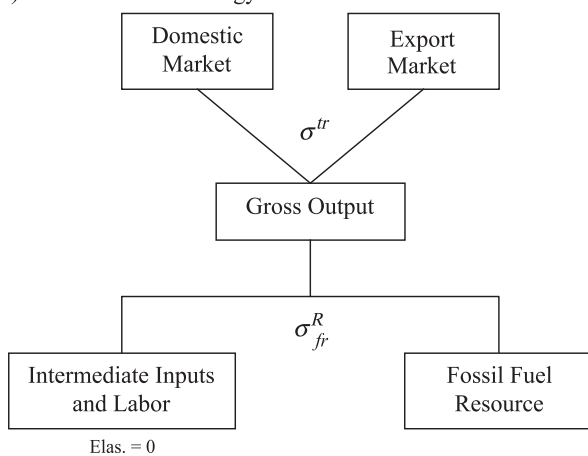


Fig. A-1. Production technologies.

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